ASISH SRI SAI KUNTA

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EDUCATION

B.S. Computer Science at University of North Texas *Expected May 2027*
Relevant Coursework: Algorithms & Data Structures, Operating Systems, Software Engineering, Systems Programming, Artificial Intelligence, Applied AI, Natural Language Processing, AI Software Development, Cybersecurity, C++.
Honors: Dean's List (Spring 2025)

PROJECTS & TECHNICAL EXPERIENCE

Elevate AI: Multi-Elevator Scheduling & Dispatch Simulator — Git Repo

- Implemented FCFS and SCAN scheduling algorithms in a pluggable Strategy-pattern architecture supporting runtime hot-swap, directly mirroring OS-level disk-scheduling strategies applied to multi-elevator resource dispatch.
- Engineered a deterministic 5-state finite state machine for each elevator (IDLE, MOVING_UP, MOVING_DOWN, PICKING_UP, DROPPING_OFF) to coordinate real-time request handling, maintain consistent state transitions, and prevent stale movement behavior across concurrent multi-elevator scheduling operations.
- Built a synchronized real-time simulation loop driving FSM transitions, request queuing, and live Pygame visualization with a metrics dashboard tracking wait time, travel distance, idle time, and efficiency score for direct algorithm comparison.
- Structured codebase into three decoupled layers (core/, schedulers/, visualization/) around a Base Scheduler interface; configured GitHub Actions CI/CD with a Python 3.8–3.11 matrix and authored pytest integration tests validating scheduler dispatch and FSM correctness.

One Step Scheduler: Full-Stack Event Management — Git Repo

- Engineered React 18 (TypeScript) frontend with React Router v6 protected routes and AUTH Context consuming a Fast API backend with a 6-table PostgreSQL schema (User, Event, Event Registration, Volunteer Assignment, Favorite, Comment) with FK constraints and cascade deletes.
- Integrated a secure JWT-based authentication and event registration workflow using pbkdf2_sha256 password hashing, HS256 token authorization, and OAuth2PasswordBearer, while supporting capacity-aware waitlisting, EUID-based ticket generation, and organizer-restricted QR check-in endpoints with role-based authorization validation.
- Contributed frontend architecture and system design artifacts (ER diagrams, API contracts) in a 5-member Agile team; connected UI to volunteer management sub-system with role-based shift scheduling and pending-approval workflows.

Campus Lost & Found - Deployed Web Application — Git Repo

- Architected and deployed a full-stack lost-and-found management platform using an Express/Node.js backend on Render, a Vercel-hosted frontend, and PostgreSQL-backed REST APIs supporting item reporting workflows, structured status transitions (reported → matched → claimed), claim management, and validated request handling for consistent data integrity.
- Implemented an automated item-matching engine comparing reports by name similarity, location overlap, and category using multi-attribute heuristic scoring to surface potential matches without manual review.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript (ES6+), Java (8/17), C++
Backend & APIs:	Fast API, Node.js, Express.js, SQL Alchemy ORM, REST API Design, JWT (HS256), OAuth2, pbkdf2
Frontend:	React 18, React Router v6, Context API, Material-UI, Axios, TypeScript
Data & Infra:	PostgreSQL, GitHub Actions CI/CD, pytest, Git, Render, Vercel, Pygame, Linux
Core Concepts:	Scheduling Algorithms (FCFS, SCAN), Finite State Machines, OOP, System Design,

CERTIFICATIONS & TRAINING

- **Oracle Java Foundations — Oracle**
  (Core Java, OOP principles, exception handling, collections framework)
- **Introduction to Generative AI — Google Cloud**
  (Generative models, prompt engineering, responsible AI)
- **Introduction to Large Language Models — Google Cloud**
  (Transformers, embeddings, LLM applications)